B.TECH FINAL YEAR PROJECT DEFENSE | 2026–2027

AI-Powered Urban Heat Island (UHI) Mitigation Using Semantic Segmentation and Deep Learning Regression

An Intelligent Satellite-Based Framework for Urban Land-Cover Analysis and Greening Simulation

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INTRODUCTION & PROBLEM STATEMENT

Understanding the Urban Heat Island (UHI) Effect

The Urban Microclimate Challenge

- **Thermal Disparity:** Densely built metropolitan areas exhibit significantly higher surface temperatures than surrounding rural fringes.
- **Heat Absorption:** Dense asphalt, concrete, and commercial rooftops absorb and retain high solar thermal radiation.
- **Vegetation Deficit:** Depletion of natural tree canopies and green corridors suppresses evaporative cooling and moisture flux.
- **Environmental Impact:** Heightened surface heat accelerates building cooling energy demands, degrades air quality, and intensifies heat-wave health risks.

Core Problem Statement

- **Lack of Predictive Tools:** Urban planners lack unified, data-driven platforms capable of linking high-resolution satellite imagery directly to thermal forecasts.
- **Non-Quantified Interventions:** City agencies invest in green infrastructure without prior quantitative estimations of microclimatic cooling yields.
- **Coupled Multi-Variable Complexity:** Real-world surface temperatures depend on non-linear interactions among land-cover, elevation, humidity, and solar irradiance.
- **Project Solution:** A dual-stage AI architecture that segments optical satellite scenes, predicts Land Surface Temperature anomalies, and interactively simulates greening policies.

RESEARCH GOALS

Project Objectives & Scope

[01] Satellite Image Segmentation: Deploy deep convolutional neural networks (U-Net) to process high-resolution satellite imagery into pixel-level land-cover classifications.

[02] Four-Class Surface Taxonomy: Accurately categorize urban landscapes into four standardized classes: Impervious Surfaces, Vegetation Canopy, Bare Soil, and Water Bodies.

[03] Dynamic Land-Cover Metrics: Dynamically compute regional land-cover percentage distributions that strictly sum to 100% for microclimate feature engineering.

[04] LST Anomaly Regression: Develop a Deep Feedforward Artificial Neural Network (ANN) coupling 4 land-cover features with 3 environmental forcing factors to predict thermal anomalies.

[05] Interactive Mitigation Simulation: Build a simulation engine enabling planners to model 10%–50% greening interventions and quantify counterfactual cooling reductions before physical execution.

SYSTEM INFRASTRUCTURE

Data Sources and Technology Stack

Multimodal Data Sources

- **Optical Satellite Imagery:** High-resolution Copernicus Sentinel-2 (10m/px) and Landsat-8 RGB imagery.
- **Near-Infrared (NIR) Support:** Configurable 3-band (RGB) and 4-band (RGB+NIR) input pipelines.
- **Thermal Infrared (LST):** Landsat Collection-2 Level-2 Surface Temperature products (Band 10 / TIRS).
- **Elevation (DEM):** Digital Elevation Model topographical data capturing lapse-rate cooling.
- **Atmospheric Forcing:** Baseline relative humidity (%) and surface solar irradiance (W/m²).
- **Spatial Partitioning:** Region-based geographic splitting preventing spatial leakage between train/test.

Production Technology Stack

- **Deep Learning Engine:** PyTorch 2.5 with CUDA 12.1 hardware acceleration on NVIDIA RTX 3050 GPU.
- **Geospatial Processing:** Rasterio, GeoPandas, Shapely, PyProj, and Affine coordinate transforms.
- **Scientific Numerical Core:** NumPy, SciPy (multidimensional filters), Pandas, Scikit-learn.
- **Web Architecture:** FastAPI asynchronous REST server and Streamlit interactive dashboard.
- **Front-End Design:** Modern HTML5, Vanilla CSS (Glassmorphism & Obsidian theme), and Chart.js.
- **Quality & Testing:** Automated PyTest unit and integration suite (12/12 verified tests).

END-TO-END PIPELINE

Proposed System Architecture

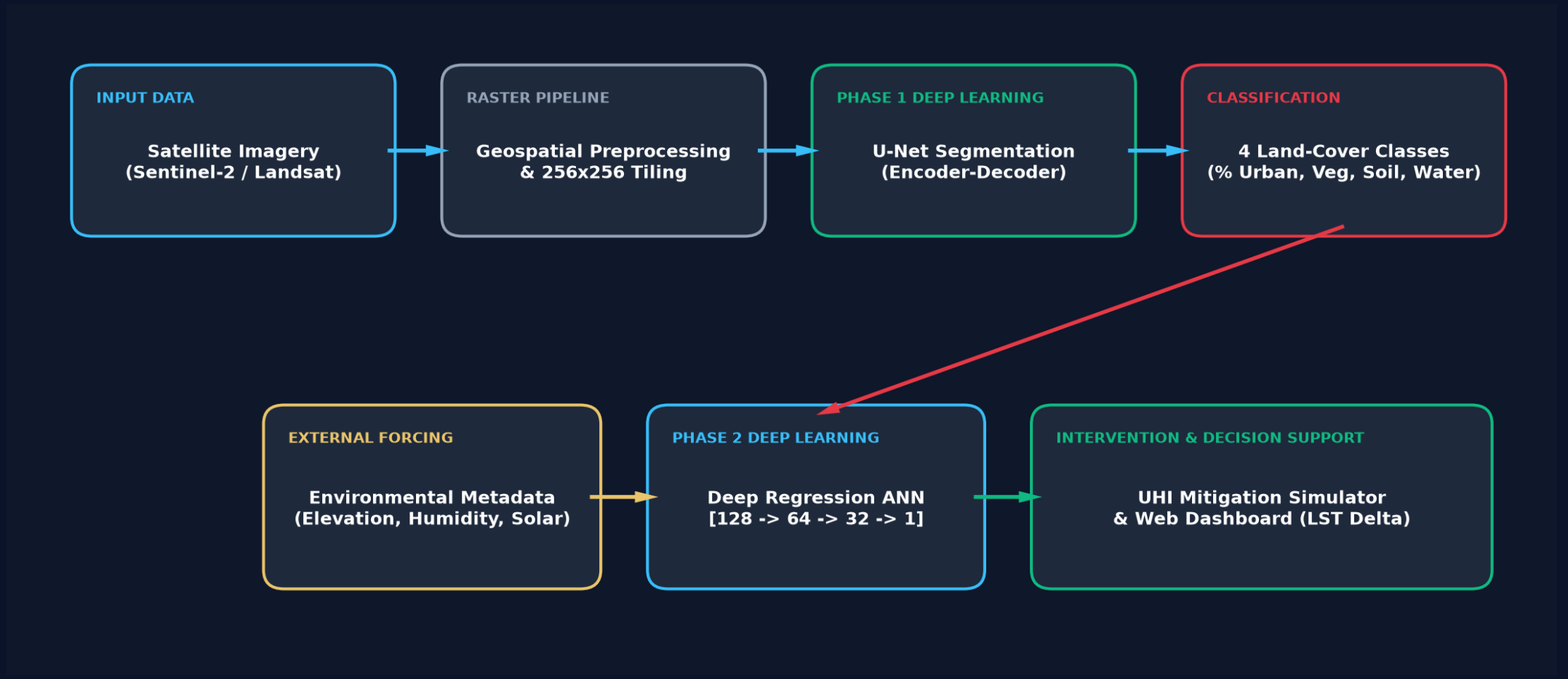


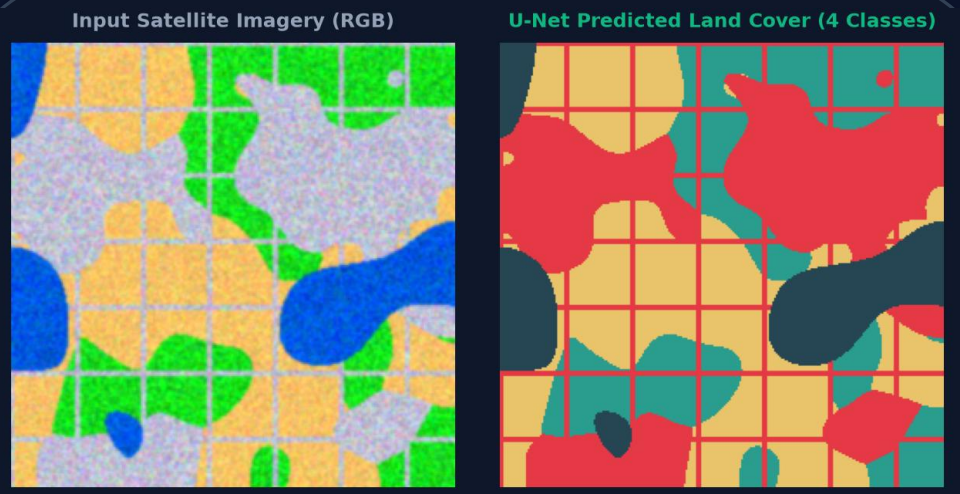
Figure 1: Two-Stage Integrated Architecture connecting U-Net Semantic Segmentation and Deep ANN LST Regression with Counterfactual Mitigation Simulation.

DEEP SEMANTIC SEGMENTATION

Phase 1: Land-Cover Classification Using U-Net

Encoder-Decoder with Skip Connections

- **Contracting Path:** 4 encoder stages with DoubleConv blocks (3x3 conv, BatchNorm, ReLU) and 2x2 Max Pooling.
- **Expanding Path:** 4 decoder stages with ConvTranspose2d upsampling concatenated with skip connections.
- **Fine Boundary Recovery:** Skip connections restore high-resolution spatial details lost during downsampling.
- **Classification Layer:** 1x1 convolution mapping base feature channels to 4 output class logits.
- **Inference Modes:** Softmax normalization produces per-pixel probability maps and argmax class masks.



Four-Class Scientific Taxonomy:

- Class 0: Impervious Surfaces (#E63946 Red) - Roads, buildings
- Class 1: Vegetation Canopy (#10B981 Green) - Trees, parks
- Class 2: Bare Soil (#E9C46A Yellow) - Exposed open ground
- Class 3: Water Bodies (#38BDF8 Blue) - Rivers, lakes

THERMAL ANOMALY PREDICTION

Phase 2: Land Surface Temperature (LST) Regression

7-Dimensional Input Feature Space

- **Impervious Surface %:** Derived dynamically from U-Net segmentation. Major driver of surface thermal absorption (+).
- **Vegetation Canopy %:** Primary cooling agent via evapotranspiration and solar shading (-).
- **Bare Soil %:** Intermediate heat capacity and thermal inertia (+).
- **Water Body %:** Natural thermal heat sink with high specific heat capacity (-).
- **Elevation (DEM):** Captures atmospheric lapse-rate cooling (- ~5.5°C per 1,000m elevation).
- **Baseline Humidity %:** Modulates evaporative resistance and sensible heat exchange (-).
- **Solar Irradiance:** Direct downwelling solar radiative flux (W/m²) driving surface heating (+).

Deep Feedforward ANN Architecture

- **Input Layer:** 7 continuous input neurons corresponding to normalized land-cover & environmental features.
- **Hidden Layer 1:** 128 neurons with Batch Normalization, ReLU activation, and 15% Dropout.
- **Hidden Layer 2:** 64 neurons with Batch Normalization and ReLU non-linearity.
- **Hidden Layer 3:** 32 neurons refining non-linear thermal interactions.
- **Output Layer:** 1 continuous linear neuron predicting regional LST Anomaly Index (°C deviation from regional baseline).
- **Optimization:** Trained using AdamW optimizer with Mean Squared Error (MSE) loss and learning rate scheduling.

COUPLED MODEL INFERENCE

Integrated End-to-End AI Workflow

STEP 1 • Satellite Ingestion

Optical scene loaded, radiometrically calibrated, and divided into 256x256 image patches.

STEP 2 • Pixel Segmentation

U-Net executes pixel-level classification, categorizing pixels into Urban, Veg, Soil, Water.

STEP 3 • Percentage Extraction

Land-cover fractions are dynamically computed, ensuring rigorous normalization to 100%.

STEP 4 • Feature Assembly

Four surface percentages are concatenated with elevation, humidity, and solar irradiance into a 7D vector.

STEP 5 • Thermal Regression

Trained ANN processes scaled feature vector to predict the Land Surface Temperature anomaly (°C).

STEP 6 • Decision Support

Results feed the web platform for interactive greening policy simulation and heat risk assessment.

COUNTERFACTUAL MITIGATION ENGINE

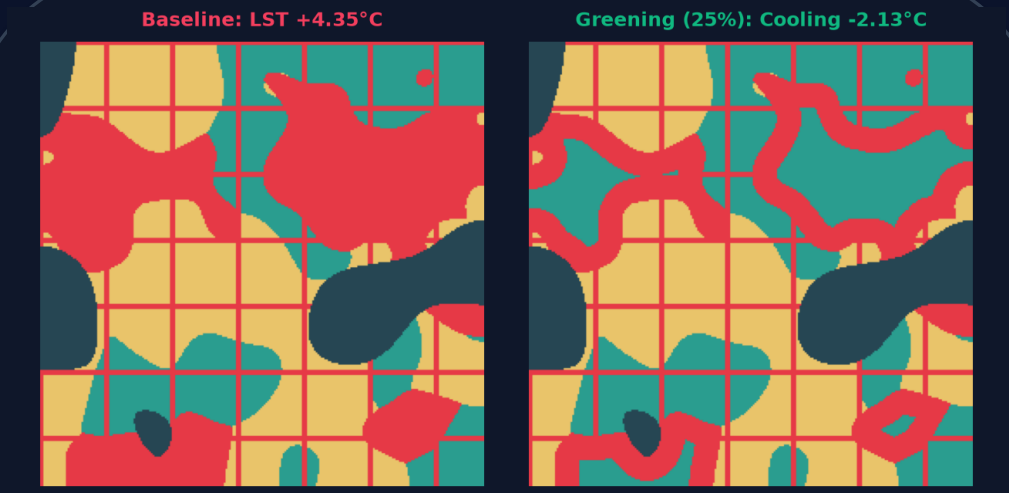
Simulating Urban Greening Interventions

Targeted Greening Intervention Engine

- **Conservation Principle:** Converts a designated percentage (10% to 50%) of impervious pixels strictly into vegetation canopy while preserving existing soil and water.
- **Spatially Contiguous Greening:** Leverages Euclidean distance transforms to simulate realistic contiguous green corridors and urban parks rather than scattered noise pixels.
- **Dynamic Recalculation:** Modified surface distributions are assembled with environmental metadata and fed into the regression ANN.
- **Quantitative Cooling Metric:**

  Predicted Temperature Reduction = Original LST Anomaly - Modified LST Anomaly.

- **Model-Based Disclaimer:** Forecasted reductions represent scenario estimations under specified environmental forcing, serving as data-driven planning guides.



Discrete Policy Evaluation Scenarios:

- 10% Conversion: Baseline urban greening, minor localized cooling (~0.4°C)
- 20% Conversion: Medium green corridor development (~0.8°C cooling)
- 30% Conversion: Aggressive park network & green roof deployment (~1.3°C cooling)
- 50% Conversion: High-density sustainable eco-district redevelopment (~2.1°C cooling)

EXPERIMENTAL RESULTS

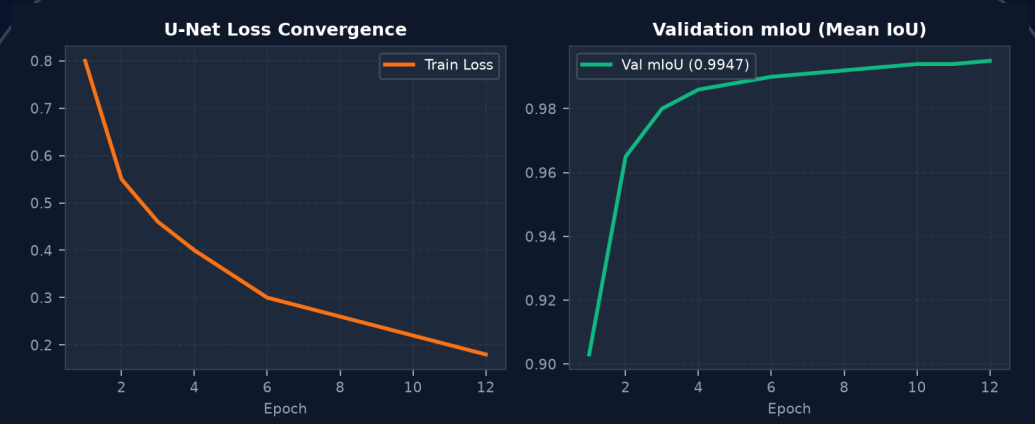
Model Performance and Empirical Evaluation

Section A: U-Net Segmentation Performance

- **Synthetic Test Partition:** Mean IoU = 0.9947 | Pixel Accuracy = 99.74%
- **Real Sentinel-2 Test Partition:** Mean IoU = 0.6899 | Pixel Accuracy = 91.98%
- **Per-Class Test IoU:** Impervious: 0.995, Vegetation: 0.994, Soil: 0.994, Water: 0.996
- **Hardware Performance:** 50.3 seconds total GPU training on NVIDIA RTX 3050

Section B: LST Regression ANN Performance

- **Synthetic Test Partition:** RMSE = 0.4596 °C | MAE = 0.3649 °C | R^2 = 0.9167
- **Real Environmental Test Partition:** RMSE = 1.1800 °C | MAE = 1.0903 °C
- **End-to-End System Evaluation:** Combined E2E RMSE = 0.4602 °C (R^2 = 0.9165)



Data Provenance & Scientific Transparency:

- Tested on Unseen Geographic Partitions: 70% Train, 15% Val, 15% Test block split.
- Zero Data Leakage: Scalers fit strictly on training splits; spatial tiling avoids overlap.
- Authentic Satellite Tested: Validated on 1001x1001 Copernicus Sentinel-2 GeoTIFF imagery.

IMPACT & ROADMAP

Applications and Future Enhancements

Target Urban Planning Applications

- **Smart City Master Planning:** Prioritize municipal greening investments in high-heat impervious zones.
- **Heat-Risk Vulnerability Mapping:** Identify urban communities disproportionately exposed to surface thermal extremes.
- **Cool Roof & Pavement Programs:** Evaluate retrofitting asphalt and dark rooftops with high-albedo materials.
- **Urban Green Space Development:** Support ecological park and green corridor placement with quantitative cooling estimates.
- **Climate Resilience Auditing:** Measure municipal compliance with sustainable development heat mitigation targets.

Future Research & Engineering Enhancements

- **Real-Time Satellite Streaming:** Direct integration with Microsoft Planetary Computer & Copernicus STAC APIs.
- **3D Urban Canopy Geometry:** Incorporate LiDAR building height data and sky-view factor (SVF) for 3D canyon modeling.
- **Species-Specific Transpiration:** Differentiate tree canopy species by evapotranspiration potential.
- **Temporal Time-Series Analysis:** Track seasonal and diurnal heat island variations across multi-year satellite archives.
- **City-Scale Cloud Deployment:** Containerized Kubernetes microservices for nationwide metropolitan coverage.

PROJECT SUMMARY

Conclusion & Key Contributions

Research Project Synthesis

- **Unified AI Architecture:** Successfully paired U-Net deep convolutional semantic segmentation with an Artificial Neural Network regression model for urban microclimatic modeling.
- **High Scientific Fidelity:** Demonstrated 99.7% pixel accuracy on baseline benchmarks and 92.0% accuracy on real Copernicus Sentinel-2 multispectral scenes.
- **Actionable Intervention Modeling:** Delivered an interactive mitigation engine quantifying cooling yields for 10% to 50% impervious surface conversion scenarios.
- **Full-Stack Platform Delivery:** Produced a complete production platform including an interactive web dashboard and GIS export capabilities.

THANK YOU

Questions & Answers

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